

Mixup Consistency Regularization for Noise-Robust Bioacoustic Classification for BirdCLEF 2026

12th place - Best Working-Note Award

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CODE / PAPER



1. TASK & CHALLENGES

1.1. Task

Multi-label bioacoustic classification in the Brazilian Pantanal

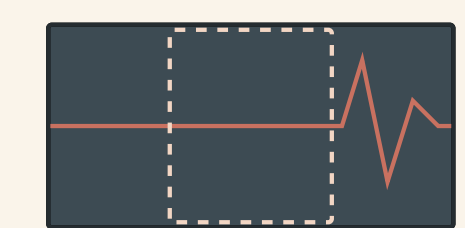
- Identify **234 wildlife species** from field soundscapes.
- Each **60-s recording** is divided into **12 consecutive 5-s segments**.
- Predict **species presence probability** for each segment.
- Hidden test set: **34% Public / 66% Private**.
- Evaluation: **macro-averaged ROC-AUC**.
- Compute constraint: **CPU-only inference**, 90 min.

1.2. Challenges



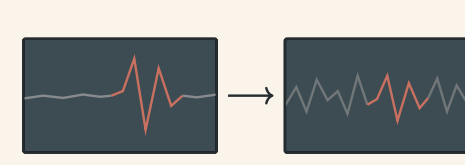
False Negatives

Background species are often audible but unannotated.



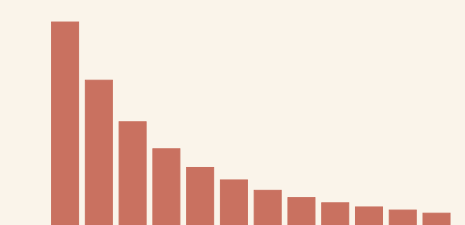
False Positives

The labeled species may be absent from a sampled 5-s crop.



Domain Shift

Clean focal recordings → noisy multi-species soundscapes.

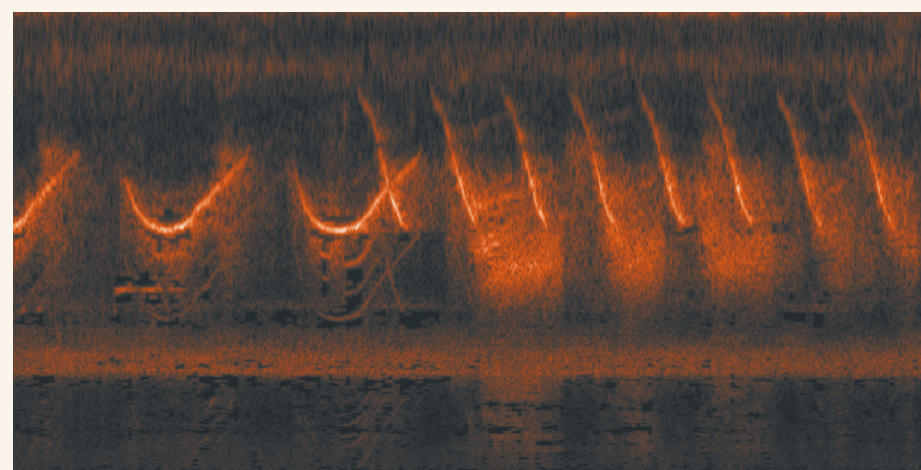


Long-tail Distribution

The number of recordings varies strongly across species.

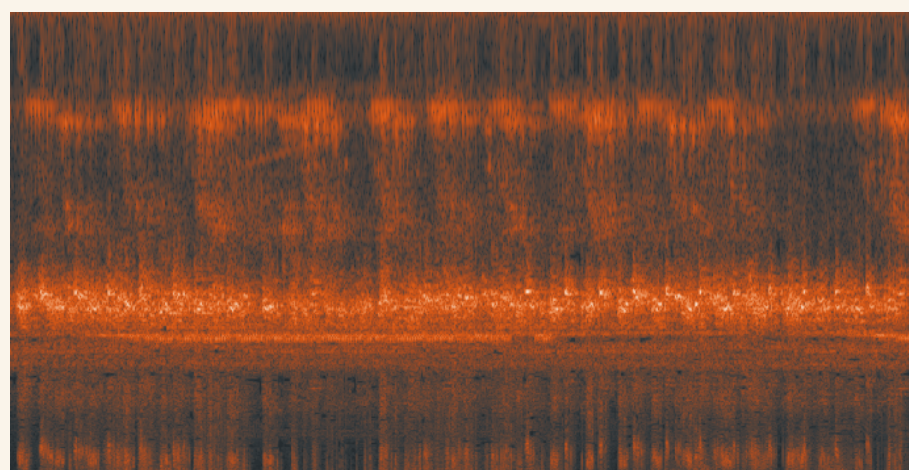
2. DATA

Focal recordings



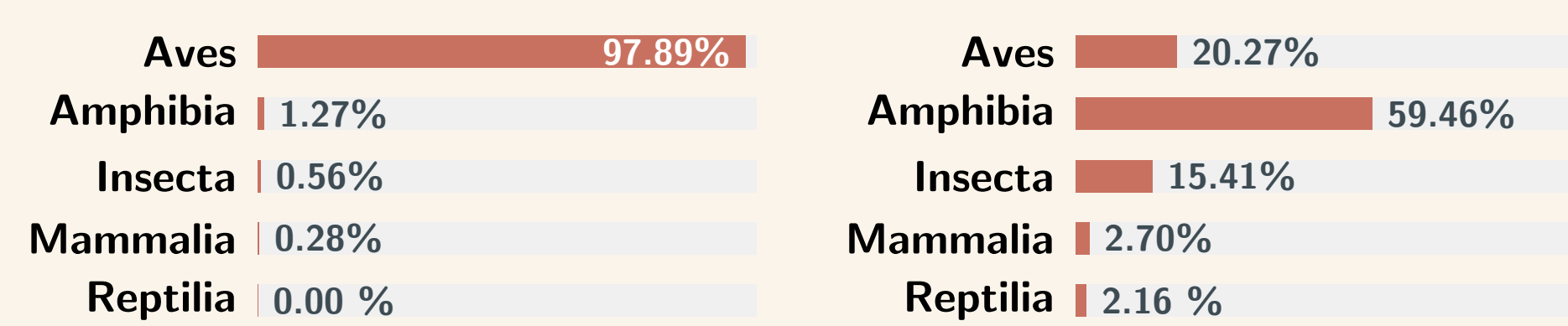
- 35,549 labeled recordings
- 206 species
- Noisy labels
- Varying recording length
- Cleaner signal

Soundscapes



- 10,658 unlabeled soundscapes
- 66 labeled soundscapes
- 75 species
- 60-s long recordings
- Noisy signal

Class distribution shift



3. VALIDATION

Focal recordings differ strongly from the test domain in both acoustic conditions and species distribution. We therefore use held-out soundscapes as our primary local validation set.

Data split

- **Focal recordings**: stratified 5-fold split by primary-label species.
- **Soundscapes**: site-aware split, accounting for species present only in annotated soundscapes.
- **Additional & silent data**: training only.

Local score = average macro ROC-AUC on held-out soundscapes across the 5 folds
Public and Private scores = scores of the ensemble of the models trained on each fold

4. METHOD

4.1. Mixup consistency

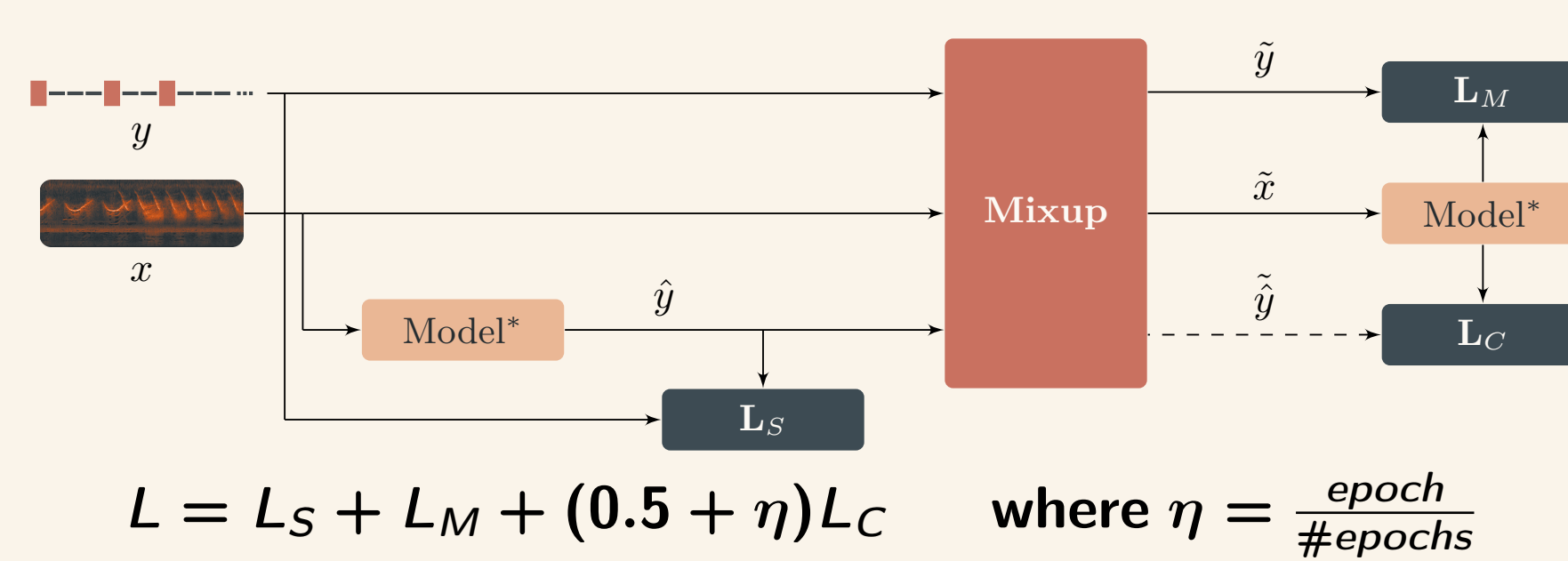
Intuition

Audio mixing is label-preserving: species detected in either source should remain detectable in their mixture, assuming no severe acoustic masking. Enforcing prediction consistency reduces sensitivity to missing labels while simulating multi-species test soundscapes.

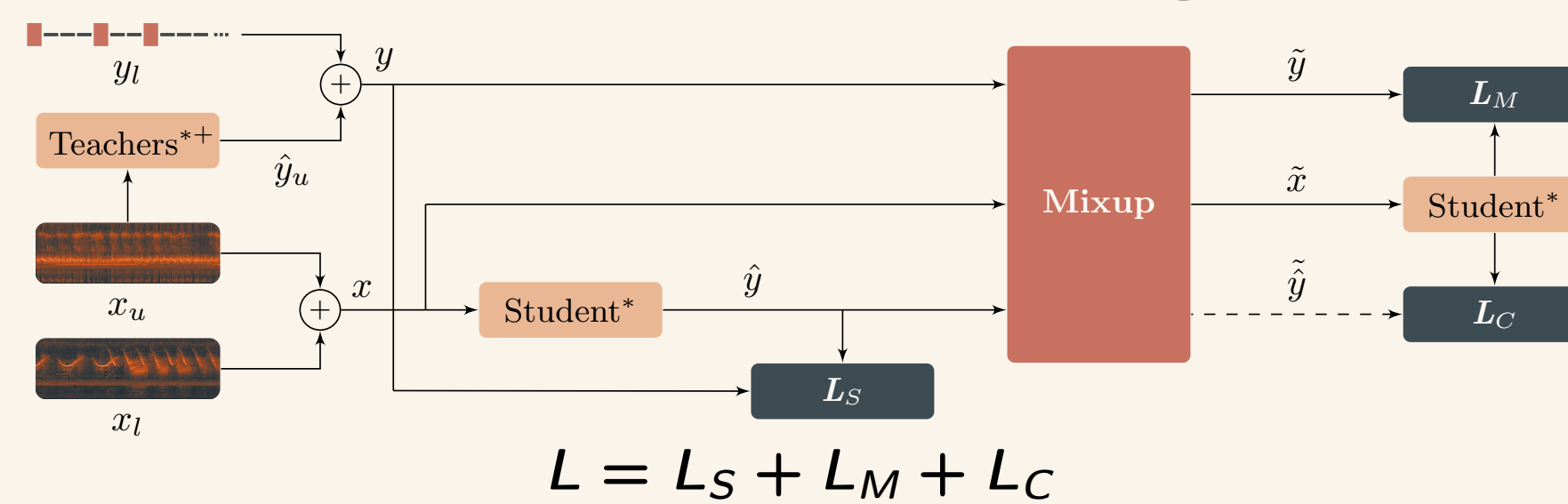
Definitions

$$\begin{aligned} \text{Mixup} \quad \tilde{x}_i &= \lambda x_i + (1 - \lambda) x_j \\ \tilde{y}_i &= \max(y_i, y_j) \\ \lambda &\sim \text{Beta}(\alpha, \alpha) \end{aligned} \quad \text{Consistency} \quad f(\tilde{x}_i) \approx \max(f(x_i), f(x_j))$$

Supervised Learning



Semi-Supervised Learning



Model setups

- **Input**: 5-s log-mel spectrograms
- **Architecture**: EfficientNetV2-B0/B3/S backbones

4.2. Class-wise Asymmetric Loss (CASL)

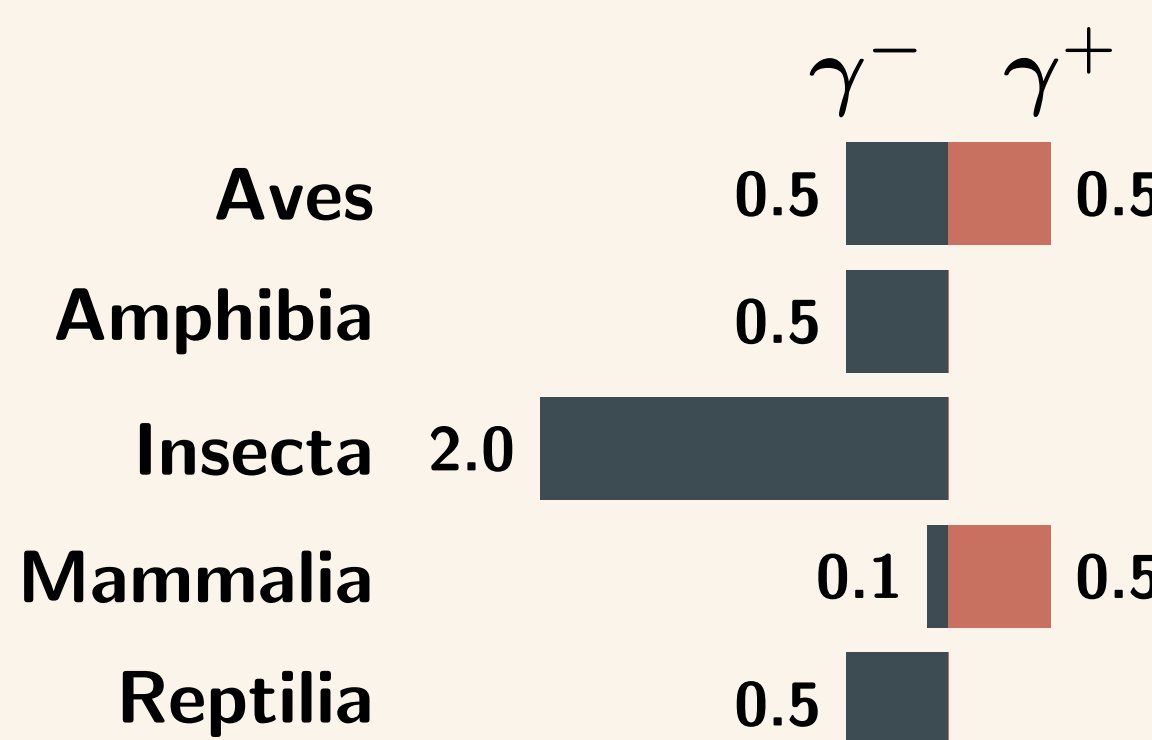
Intuition

Annotation noise is class-dependent: taxa differ both in their likelihood of being present but unlabeled in the background, and in their likelihood of being cropped out of a sampled 5-s window due to differences in call duration and frequency.

Definitions

$$\begin{aligned} \text{Asymmetric Loss (ASL)} \quad L &= -y L_+ - (1 - y) L_- \\ \text{ASL} &= \begin{cases} L_+ = (1 - p)^{\gamma^+} \log(p) \\ L_- = (p_m)^{\gamma^-} \log(1 - p_m) \end{cases} \\ p_m &= \max(p - m, 0) \end{aligned}$$

Larger γ^- reduces the influence of unreliable negative labels, while γ^+ controls the contribution of potentially unreliable positives.



5. RESULTS

5.1. Main ablation

Loss	Consistency	Soundscape ROC-AUC	Public	Private
BCE	—	0.900 ± 0.041	0.926	0.931
BCE	✓	0.894 ± 0.033	0.924	0.932
CASL	—	0.898 ± 0.036	0.925	0.931
CASL	✓	0.918 ± 0.024	0.938	0.940

CASL and Mixup consistency are complementary rather than independently effective. Neither component improves performance alone, while their combination yields a clear gain across local validation and both leader-board splits.

5.2. Pre-training

Pre-training	Soundscape ROC-AUC	Public	Private
—	0.901 ± 0.026	0.928	0.927
✓	0.918 ± 0.024	0.938	0.940

Bioacoustic pre-training consistently improves performance, suggesting that domain-specific representations transfer substantially better than ImageNet initialization.

The pre-training stage consists of training a backbone on genus-level classification over 296,052 recordings from Xeno-Canto.

5.3. Semi-supervised learning

Stage	Public (up to)	Private (up to)
Supervised (teachers ensemble)	0.938	0.951
Semi-supervised (student)	0.950	0.952

Pseudo-labeling unlabeled soundscapes produces stronger models overall, indicating that training on genuine soundscape signals helps reduce the focal-to-soundscape domain gap.

5.4. Post-processing

	Public	Private	ΔPublic	ΔPrivate
Ensembling	0.948	0.941		
+ Logits sharpening	0.948	0.942	+0.000	+0.001
+ Sliding window	0.956	0.952	+0.008	+0.010
+ Smoothing	0.959	0.956	+0.003	+0.004
+ Priors	0.960	0.956	+0.001	-0.000

Sliding-window inference provides the largest post-processing gain, while temporal smoothing brings an additional improvement. Metadata priors have only a marginal effect.

5.5. CASL parameters ablation

Class	Soundscape ROC-AUC	Public	Private
Aves	0.908 ± 0.025	0.922	0.933
Amphibia	0.901 ± 0.039	0.930	0.936
Insecta	0.920 ± 0.034	0.935	0.941
Mammalia	0.903 ± 0.030	0.935	0.937
Reptilia	0.912 ± 0.021	0.932	0.936

To isolate the effect of class-wise negative focusing, we apply $\gamma_- = 2$ to one taxonomic class at a time and set the other parameters to 0. **The strongest improvement occurs for Insecta**, supporting the use of taxon-specific loss parameters to account for different levels of missing-label noise.

TAKEAWAY

Consistency regularization and CASL are most effective together. Mixup consistency seems to provide a useful self-supervised signal only when unreliable class-dependent gradients are sufficiently attenuated.

WHY IT MATTERS

Noisy labels and domain shift can be addressed jointly rather than independently. Exploiting the label-preserving nature of audio mixing lets the model learn from its own predictions while simultaneously exposing it to synthetic multi-species soundscapes.

LIMITATIONS / FUTURE WORK

- CASL parameters were chosen from qualitative assumptions, not estimated noise rates.
- The proposed explanation for the synergy between CASL and Mixup consistency remains a hypothesis.
- Further pseudo-labeling iterations did not improve performance, suggesting confirmation bias remains an issue.